ZeroBreach AI — Enterprise Technical Architecture & System Specification

Platform Version: 2.5.0-Production | Engine: Dual-Model Ensemble (Random Forest + XGBoost) | Classification: Enterprise NDR/NIDS

1. Executive Overview

ZeroBreach AI is an enterprise-grade Network Intrusion Detection & Threat Intelligence Platform (NIDS/NDR) engineered to ingest, inspect, and evaluate network packet flows in real-time. By bridging high-throughput packet sniffer hardware/software layers with machine learning inference engines, ZeroBreach AI flags malicious anomalies, calculates context-aware risk scores, profiles targeted infrastructure assets, and arms Security Operations Center (SOC) personnel with operational telemetry.

2. User Roles, Personas & Access Control (RBAC)

Role	Access Tier	Allowed Capabilities	Restricted Endpoints
Administrator (admin)	Full Root	User Management, API Tokens, Audit Logs, DR Backups, SIEM, System Config	None (Complete unrestricted access)
SOC Lead (soc)	Operational Command	Live Packet Sniffing, PCAP Ingestion, Threat Triage, SIEM Export, Tickets	User Role Creation, Root Secrets
Security Analyst (analyst)	Core Telemetry	PCAP Upload & Ingestion, Threat Hunting, Live Sniffer Monitoring, Ticket Submissions	Global System Configuration, DR Snapshots
Enterprise User (enterprise)	Executive Read-Only	3D Telemetry Dashboard, Threat Frequency Spectrum, Protocol Sphere, SLA Reports	PCAP Upload, Sniffer Start/Stop, SIEM Export

3. Dual-Model Machine Learning Engine (Champion-Challenger Architecture)

To eliminate single-model bias, minimize false positives, and maximize intrusion detection recall on subtle zero-day vectors, ZeroBreach AI operates a Dual-Model Ensemble Inference Engine featuring real-time Dynamic Confidence Arbitration.

Characteristic	Model 1: Primary (Champion)	Model 2: Secondary (Challenger)
Algorithm Family	Random Forest Classifier (RandomForestClassifier)	Gradient Boosted Decision Trees (XGBClassifier)
Ensemble Paradigm	Bagging (Bootstrap Aggregation of Deep	Boosting (Sequential Residual Error

	Trees)	Optimization)
Estimator Capacity	100 Trees (n_estimators=100)	100 Gradient Boosted Trees (max_depth=6, lr=0.1)
Feature Dimensionality	81 Statistical NetFlow Features	81 Statistical NetFlow Features
Target Classes	13 Attack Classes (CICIDS2017)	13 Attack Classes (CICIDS2017)
Specialized Strength	Broad boundary generalization, noise resistance	Complex multi-feature edge cases & rare threats
Arbitration Method	Dynamic Confidence Maximization (Winner: max prob)	Dynamic Confidence Maximization (Winner: max prob)
Disagreement Logging	Tagged in MongoDB traffic_logs (model_disagreement: bool)	Tagged in MongoDB traffic_logs (model_disagreement: bool)

Dynamic Confidence Arbitration Algorithm

1. Both models evaluate the 81-feature flow vector in parallel.
2. Winning class confidence is calculated: c_RF = max(P_RF) and c_XGB = max(P_XGB).
3. Winner Selection: The prediction from the model with the higher class confidence is chosen as the final alert.
4. Disagreement Auditing: If the two models predict differing classes, model_disagreement is flagged as True and stored in MongoDB.

4. Context-Aware Risk Scoring Mathematical Formula

Final Risk Score = max((W_conf * C) + (W_sev * S) + (W_asset * A)) * 100, Context Floor)

Where:
• W_conf = 0.5 (50% weight): Model Confidence (max predicted class probability)
• W_sev = 0.3 (30% weight): Attack Category Severity (0.30 for Portscan up to 1.00 for Infiltration/Heartbleed)
• W_asset = 0.2 (20% weight): Targeted Asset Criticality (0.80 for App Servers, 1.00 for Production DB/Domain Controller)
• Context Floor = 60.0: Enforced whenever attack severity >= 0.95 and asset criticality >= 0.80.